

Do longer housing searches lead to better matches? Evidence from the voucher program*

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Abstract

Low-income households often struggle to find affordable housing that meets their needs. While the Housing Choice Voucher program helps over 2.3 million households per year afford rental housing, participants in the program still face heightened challenges in using their vouchers, finding stable housing, and reaching higher opportunity neighborhoods. In this paper, I investigate the role of search time in promoting better housing matches within the voucher program. Using features of the random timing of voucher issuance, I show that there is no meaningful effect of issuance timing on the probability of using a voucher (in other words, the probability of making any match), but there is a strong effect on the length of time that households search for when they do use their vouchers. I then use issuance timing as an instrument for search time, and find that longer searches are associated with declines in housing turnover; specifically the probability of exiting the program or moving housing units within 2 years of entering. Longer searches also allow households to reach lower poverty neighborhoods than baseline. Effects on housing turnover persist across market conditions, but effects on reaching lower poverty neighborhoods are driven by tighter housing markets, where search costs may be highest.

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1 Introduction

In increasingly tight housing markets, low-income households often struggle to find safe and affordable rental housing that meets their needs. Due to a combination of rising rents, stagnating incomes, and a lack of new affordable homes, more than half of all US renters were cost burdened in 2022, paying more than 30 percent of their income in rent, an all time high (Joint Center, 2024). There are certain groups in particular that face heightened challenges in the housing search process, including Black and Hispanic households (Christensen et al., 2021; Christensen and Timmins, 2021; Arefeva et al., 2024), and households with children (Aron et al., 2016), who have been shown to face heightened discrimination in the housing search process.

Another group that may struggle to find housing is voucher recipients. The Housing Choice Voucher program subsidizes over 2.3 million households annually; however, voucher recipients have to find and rent homes on the private rental market to receive their subsidies. In addition to contending with a shortage of affordable homes, voucher recipients have several additional challenges. First, they have a limited time to search for housing in order to receive their benefits. They also must find housing units that adhere to all program guidelines. Voucher recipients also must find a landlord who is willing to rent to them and participate in the program, which can be a high hurdle to clear as landlords often discriminate against voucher holders (Aliprantis et al., 2022; Faber and Mercier, 2022; Cunningham et al., 2018; Freeman and Li, 2014; Phillips, 2017).

The consequences of the difficult housing search process are reflected in programmatic outcomes. Most fundamentally, in recent years, less than 60% of households issued a voucher are able to use that voucher at all. Even for households who are successful in using their vouchers, there is still evidence of high search costs, which can be measured by looking at search time, or the elapsed time between the issuance of a voucher and a participant successfully using their voucher to lease a home. Search times increased sharply between 2018 and 2022, now standing at around 84 days, which is considerably longer than the federal minimum 60 days that housing authorities must give households to use their vouchers. Additionally, despite the voucher program’s core programmatic goal of opening up more neighborhoods to program participants, households with vouchers often live in similar neighborhoods to unassisted, low-income renters (Jacob et al., 2015; Galvez, 2011; Pendall, 2000). Low success rates, growing search times, and constrained locational choices can all

be reflections of the difficulty of the housing search process for voucher recipients.

In this paper, I study the effect of search time on housing match quality in the voucher program. Though there is wide variation in search times both across and within the 2,000+ housing authorities who administer the program across the country, there is very little research that looks at the drivers of this variation or the effects of longer searches. Specifically, I evaluate how longer searches can lead to lower housing turnover, in terms of program exits and moving units, and the ability of households to reach lower poverty neighborhoods.

I use novel administrative data to measure success rates and search times for over 700 housing authorities across the country, which allows a more detailed analysis of these programmatic outcomes than past studies. The empirical strategy is derived from the fact that the timing of voucher issuance is random, but there are features of issuance timing that can affect the housing search process. I focus on the month of the year that a voucher is issued, which can create variation in search outcomes based on both seasonality in the rental market, and how PHAs may administer and enforce search time policies across the calendar year. This source of random variation in search time allows for the causal identification of the effect of longer searches. I also test an additional instrumental variables strategy that leverages the timing of issuance within a month.

First, I test the effect of the month of voucher issuance on the probability of successful voucher use- in other words, the effect on making any match. Households who are issued a voucher in January are marginally more likely to successfully use their voucher; throughout the rest of the year, the probability of successful voucher use is relatively stable. Given the minimal effect on success rates, but more significant effects on the length of time that a successful households searches for, I then use the month of issuance as an instrument for observed search time to test the effect of longer searches on several different margins of match quality. I also conduct heterogeneity tests, motivated by the fact that searches tend to be longer for certain groups, including households of color, and are considerably longer on average in tighter markets.

I find that longer searches lead to increased program retention, and a longer lengths of stay in the same housing unit. Specifically, a 50 percent increase in search time leads to 2.3 percentage point (12%) percent decline in leaving the program within 2 years, and a 3.5 percentage point (20%) decline in moving units within two years. Declines in housing unit turnover are most pronounced for Black households and younger households, and persist across rental market conditions.

Longer searches also allow voucher recipients to reach lower poverty neighborhoods. Increasing search time by 50 percent leads to a 2.5 percentage point (7%) increase in the chance of moving to a ZIP Code with a lower poverty rate than baseline. These results are predominately driven by Black households and households with kids, and, unlike the effects on housing turnover which are prevalent across housing market characteristics, are entirely driven by tighter rental markets.

This paper contributes to several strains of literature. First, these results draw important parallels to other literature on search and match quality. Perhaps most closely related is the job search literature, where several studies show that longer job searches can lead to better matches, predominately measures by longer stays (Belzil, 2001; Caliendo et al., 2013; Ehrenberg and Oaxaca, 1976).

Second, this paper contributes to the literature on housing search. While there is a large body of work on housing search, much of the quantitative evidence focuses on the home-buying process. Several studies show differences in the housing search process by income, race, and type of constraints (Harvey et al., 2020; Krysan and Crowder, 2017). This paper also adds to a body of predominately qualitative work on the search process for voucher recipients (Garboden, 2021; McCabe, 2023; Rosen, 2014, 2020), and a small but growing set of work on success rates and drivers of success rates within the voucher program (Ellen et al., 2024b; Chyn et al., 2019; Finkel and Buron, 2001). The results on neighborhood quality also tie into research on housing mobility programs, revealing a potentially low or no-cost method of expanding neighborhood choice in the voucher program (Bergman et al., 2024; DeLuca et al., 2023).

Third, this paper adds to a growing literature on administrative burden, and how features of a program’s design can affect take-up and retention. Homonoff and Somerville (2021) show that the timing of certain milestones within a program can influence program retention. Several other papers show that features of program design can affect take-up and other programmatic goals through the administrative burden channel (Gray, 2019; Ganong and Liebman, 2018). This paper adds to this literature by providing new evidence on a potential strategy (extending search times) to boost take-up and retention in a program that involves a complex interaction with a private third party.

The paper is organized follows. Section 2 details the institutional context, including background on voucher success rates, the housing search process within the voucher program, and the theoretical

argument on how longer searches can affect different margins of match quality. Section 3 describes the data used for the study. Section 4 provides descriptive information on search and matching within the voucher program, voucher issuance trends, and allowable search time policies. Section 5 describes the empirical design and Section 6 presents causal results on the effect of issuance timing on success, and the effect of search time on housing turnover and neighborhood outcomes. Section 7 discusses robustness, and Section 8 concludes.

2 Background

2.1 Background on the voucher program

The Housing Choice Voucher program is the largest federal rental assistance program in the country. In 2022, more than 5 million individuals in 2.3 million households received vouchers. While the program is funded federally by the Department of Housing and Urban Development (HUD), it is administered locally by more than 2,000 public housing authorities (PHAs). PHAs have discretion over the daily operation of the program, but abide by certain federal guidelines set by HUD.

Vouchers can be transformative for low-income households who are able to use them. Once a household enters the voucher program, they generally pay 30 percent of their income towards rent, and the local housing authority covers the rest, up to a certain subsidy limit, called the payment standard. Rent burdens in the voucher program are significantly lower than those of unassisted, low-income renters (Ellen and Torrats-Espinosa, 2020). Vouchers have also been shown to reduce homelessness, reduce over-crowding, and improve children’s performance in school (Jacob and Ludwig, 2012; Mills et al., 2006; Wood et al., 2008; Schwartz et al., 2020).

However, the program struggles with two main issues. This first is that many households who receive vouchers are not able to use them. In recent years, as much as 40 percent of households who receive vouchers must return them to the housing authority as they are unable to find and rent a qualified home within the allotted time they are given. Second, the voucher program has not consistently achieved its core goal of allowing voucher holders to live in a diverse set of neighborhoods (Galvez, 2011; Pendall, 2000). Both of these issues can reflect the difficulty that voucher recipients may have during the search process.

2.2 Take up in the voucher program

Take-up in the voucher program is a multi-step process, often spanning multiple years. First, voucher holders must apply to join a waitlist at a PHA. Many PHAs will close their waitlists due to outsized demand for the program, so in order to even apply, participants must often wait for the PHA in their local market to open their wait list (Hembre and Urban, 2023). At this point, households apply, and if they are certified to qualify for the program, they are placed on the wait list. While wait times vary widely across the country, they are generally quite long. On average, households wait 2.5 years, but at some housing authorities included in the sample for this paper, wait times are as high as 8 years (Acosta and Gartland, 2021).

Once a household is selected off the waitlist, they are contacted by the PHA, confirm their eligibility and interest in the program, and provide documentation. Only then are they issued a search voucher. A search voucher is a promise of receiving the subsidy, given the voucher recipient is able to secure a qualifying unit. At this point, households must be given a minimum of 60 days to search, though many PHAs offer longer search windows. PHAs also have discretion in granting extensions. Many PHAs offer some sort of automatic extensions, and virtually all PHAs grant extensions for a wide variety of acceptable reasons. In the voucher program guidebook, HUD notes that search time policies should be monitored and evaluated, to determine how allowable search time may affect leasing success; to my knowledge, this is the first paper that addresses either allowable or observed search times in the voucher program.

Households now must find a unit that meets all programmatic requirements. The unit must have a rent that falls within the payment standard, and must pass a housing quality inspection.¹ The landlord for the unit must also be willing to rent to a voucher holder and participate in the program.

Vouchers are not an entitlement, which makes calculating a traditional take-up rate difficult. Only one in five low-income households with unmet housing needs get any form of housing assistance (Kingsley, 2017). In this paper, I focus only on eligible households who are issued a voucher, a small subset of all eligible households. The share of households who are able to use their vouchers once

¹Payment standards are based on HUD's Fair Market Rents (FMRs). PHAs have discretion to set payment standards within 90-110% of FMRs. Voucher holders may rent units above the payment standard, but they are required to pay the difference between the rent of the unit and the payment standard out-of-pocket, and when a voucher recipient joins the program, their rent payment cannot exceed 40%.

they are issued a voucher is called the success rate; a search is considered successful if the household is able to use the voucher within one year of issuance.² It's important to note that at this point in the process, households have repeatedly expressed interest in the program and confirmed their eligibility. The voucher program also provides quite generous subsidies. The average household in the sample for this paper received \$8,000 per year, and in more expensive housing markets, this number can be substantially higher. This makes low, and declining, success rates even more striking, and invites further examination of the housing search process.

2.3 Factors affecting the search process

In general, low-income households often face a difficult housing search process. Evidence shows that low-income households who are more constrained in their search often have different goals from higher income households; while higher income households search for "forever homes" that will meet their long-term needs, lower income households often end up accepting homes that they view as "temporary stops" and sacrificing on some of their long term goals and preferences (Harvey et al., 2020). These differences can often be exacerbated further when households are forced to move, which is particularly relevant for new voucher recipients who have only a limited time to use their voucher.

The voucher program predominately assists Black and Hispanic households. Around 50% of voucher households contain children, with over 90% of those households headed by a single mother. The demographics of the voucher program mean that participants may face confounding challenges. Households of color have been consistently shown to face discrimination in both rental and homeownership markets (Christensen et al., 2021).

In addition to the challenges discussed above, voucher recipients have additional requirements that they have to navigate. While they do not face a traditional budget constraint (voucher holders are theoretically indifferent on rent, up to the applicable payment standard), they have to meet several other conditions that can cause many potential voucher holders to either fail to use their voucher altogether, or fail to meet some of their preferences. The first is finding units that are eligible for the program. Cunningham et al. (2018) find that across 5 cities, only 1 in 39 rental

²The one year window allows for comparable measures across years. Most searches that are successful will conclude within a year; in the sample for this paper, 99 percent of successful searches ended within a year.

units were eligible for the program, given subsidy limits.

The second is finding a landlord that will accept vouchers. Several studies have documented both direct and indirect discrimination against voucher holders. Source of Income (SOI) discrimination laws prohibit landlords from discriminating against potential tenants on the basis of using a voucher. While many states and localities have enacted such laws, as of 2022, only about 57% of voucher holders live in jurisdictions with SOI protections (Knudsen, 2022), and there is still frequent discrimination against voucher holders even where SOI laws are present (Cunningham et al., 2018; Faber and Mercier, 2022). Landlords are often hesitant to house voucher recipients both because of preconceived notions about voucher recipients themselves, and due the added administrative burden of complying with program requirements, passing housing quality inspections, and working with housing authorities (Garboden et al., 2018). Blanco and Song (2024) find that landlords in higher rent areas are more likely to use negative language towards voucher holders when advertising their units, indicating that it might be even more difficult to find landlords who will accept vouchers in a broader set of neighborhoods. Research showing that voucher recipients reach lower poverty neighborhoods where SOI laws are present also provides evidence of the barriers that landlords can impose on the locational choices of voucher recipients (Freeman and Li, 2014; Ellen et al., 2022).

Voucher recipients may also struggle with the costs associated with administrative burden. Learning, compliance, and psychological costs may all be at play during the housing voucher search process (Herd and Moynihan, 2018). Voucher recipients have only a short time to learn about all program requirements, and understand what these requirements mean in their local context. And although at this point in the process, voucher recipients have already confirmed their eligibility, there are still several administrative steps they must take to find an eligible unit, get it approved, and sign a lease. Psychological costs may also affect the housing search, both due to fear of rejection from landlords in a wider set of neighborhoods (DeLuca et al., 2023), stigma associated with using a voucher, and the difficulty of moving further from existing homes and social networks.

2.4 The role of search time

There are several reasons to believe that voucher recipients who search for longer may increase the chance of successfully using a voucher, and the quality of the resulting match for successful households. In the literature on job search, a common finding is that longer searches can lead to

a better match, most commonly measured by the length of stay at the next place of employment (Belzil, 2001; Caliendo et al., 2013; Ehrenberg and Oaxaca, 1976). The theory underlying these models is that given a shorter time constraint, workers may accept a job that they would otherwise reject, had they been given more time to search. This is frequently evaluated in the context of unemployment insurance, where an important feature of policy design is the benefit duration. This has strong parallels to the search process within the voucher program, where recipients are given a limited duration of time to search for a unit.

Longer searches may also allow participants more time to expand their search beyond their immediate neighborhood and social networks.³ Search costs can be substantial; perhaps particularly for the large share of voucher households that contain school-age children. Additional flexibility in timing can make it possible for households to search outside of their neighborhood, and ensure that they are able to make the move to a new area within the allotted time.

Additional time can also provide voucher recipients with a better understanding of program requirements, helping to combat the high compliance and learning costs that the voucher program entails. Several papers show that compliance and learning costs can depress program take-up across other government programs, and allowing for additional time and support in navigating these costs can have large benefits (Homonoff and Somerville, 2021; Chetty et al., 2013; Bettinger et al., 2012; Barr and Turner, 2018; Armour, 2018).

3 Data

3.1 Administrative data on voucher use

I use detailed administrative data from the Office of Public and Indian Housing (PIH), the division of HUD that is responsible for administering the voucher program. PHAs submit data on each programmatic action to the PIH Information Center (PIC). These actions indicate when each household is issued a voucher, and if and when they are successful in using their voucher. Importantly, I observe key demographics, including race, ethnicity, age, household composition, and origin ZIP Code at the time of voucher issuance. For households who are successful, once they

³Social networks have shown to be an important force in the residential choice of voucher holders (Ellen et al., 2019).

are admitted to the program, I observe their full demographics, complete address, and detailed information on their housing costs and subsidy payments.

For households who are successful in using their vouchers, I also observe all subsequent actions, including moves, ports to other PHAs, income re-certifications, and program exits. From these additional actions, I can construct a history of program participation, and identify how long they remain in the voucher program and at a specific address.⁴

Neighborhood outcomes are derived from merging this administrative data with data from the 2015-2019 American Community Survey (ACS) on poverty rates and poverty percentiles within a CBSA at the ZCTA level. I identify a dichotomous indicator for moves where the destination poverty percentile is higher than the origin poverty percentile; I also use a continuous indicator for the change between the origin and destination poverty percentile.

I use several supplementary data sources to conduct heterogeneity analysis. I use county rent levels from the 1-year 2019 ACS to stratify the sample into low and high rent markets. I also use data on vacancy rates from HUD and the United States Postal Service as an additional method for stratifying the sample based on rental market conditions.

3.2 Sample

The administrative data greatly expands the ability to measure success rates and search times. However, these data have not historically been used to measure these outcomes, and required extensive cleaning. In order to get accurate estimates of these two metrics, I employ data quality standards that select only PHAs where data on voucher issuance appears to be complete (Ellen et al., 2023). More information about these quality standards can be found in Appendix A.

The sample for this paper includes 722 PHAs from the 300 largest CBSAs, in terms of number of voucher holders.⁵ Combined, these PHAs cover nearly 55% percent of voucher holders. In general, these PHAs are very reflective of the program as a whole (appendix table A1).

Table 2 reveals important characteristics of the sample. Just over half of households are Black; around 15 percent of households are Hispanic. Half of household include children, with the large

⁴Specifically, I detect changes in addresses by geocoding each unique address entry. The geocoding match rate is above 95%. I then use the cleaned addresses to detect any changes in address. This methodology ignores moves within the same building, so a move is defined as a change of street address, not simply a change in unit. Changing units within a building is relatively rare; I estimate that only around 3% of moves are within the same building.

⁵I exclude 39 legacy Moving To Work (MTW) PHAs, as these PHAs do not report the ZIP Code of origin.

majority (90%) of households with kids headed by women.

3.3 Originally collected data from administrative plans

There is no centralized source of information on allowable search time policies by PHA. PHAs record these policies in annual Housing Choice Voucher Administrative Plans, which do not receive formal approval from HUD, so records of these plans are not maintained. I collect administrative plans from several sources. First, historical plans for 271 PHAs in my administrative data sample were provided by Zhang and Johnson (2023), who hand-collected plans from PHA websites and outreach between 2016 and 2020. An additional 32 plans were obtained from the Boston Regional HUD office via a Freedom of Information Act request. Finally, additional plans were collected directly from PHA websites; however, plans collected from websites generally cover policies going into effect in 2023; the administrative data for this paper goes through 2022 only. An additional 23 plans from the years 2015-2022 were gathered directly from PHA websites.

From the administrative plans, I record data on the initial search term, automatic extensions, discretionary extensions, tolling and expiration practices, and any exceptions that PHAs make in search times. I then classify PHAs by their "effective" allowable search time, which is their initial search time plus time given from any automatic extensions. More information about the data collection process can be found in Appendix [B](#).

4 Descriptive facts about success rates and search times

This paper is one of the first in a growing body of work on success rates and search times in the voucher program (Ellen et al., 2024b). In this section, I document several stylized facts about success rates and search time that inform the causal analysis that follows, including recent trends, differences across PHAs, demographics, and markets, seasonal patterns, and early evidence on allowable search times across PHAs.

4.1 Trends in success rates and search times

Low success rates have long been a key challenge for the voucher program, but until recently there was very little insight into voucher success rates or search times. The administrative data

used for this paper is one of the only methods that allows for measuring both how many households are able to use their vouchers, and how long this process takes. In this section, I present several descriptive facts about success rates and search times that motivate the following causal analysis.

Prior to 2021, success rates (defined as the share of households who are able to use their voucher within 1 year of issuance) hovered around 60%, but have since declined sharply. By 2022, just over 55% of those issued a voucher were able to lease up and enter the program. At the same time, search times have been sharply increasing. In 2022, it took the median voucher recipient around 75 days to use their voucher, up more than two weeks from two years prior. Both of these metrics indicate growing search costs for voucher recipients.

There is also wide variation in success rates and search times across the country (Figure 2). Ellen et al. (2024b) explains some of this variation with a combination of individual characteristics of households, characteristics of origin neighborhoods where searches originate, and market conditions, but a large share of this variation still remains unexplained. Differences in search time are particularly large, which can be due to differences in PHA policies, differences in market conditions, and how PHAs respond to differences in market conditions. We see evidence of this in appendix figure C2 as well, which shows that searches are considerably longer in tighter rental markets, measured by median county rents.

There are also large differences in search times by race and ethnicity, and these difference persist even when we take into account differences in household composition. The median Black household searches for 25 days longer than the median white household (Table 2). Within the voucher program, older households tend to have shorter search times, as they are often less likely to move from their initial unit. But even when we take into account differences by age, Black households still search for longer; the median Black voucher recipient over the age of 65 searches considerably longer than the median White household under 65 (appendix figure C1). Searches also tend to be longer in housing authorities in higher rent counties (appendix table C2).

In Table 3, I show raw means for the dependent variables on housing turnover and neighborhood outcomes. Longer searches have lower rates of exiting and moving within 1 and 2 years. Moving to a lower poverty neighborhood shows a similar trend, though not monotonic and slightly less pronounced. Interestingly, we also see that longer searches are associated with moving to a higher poverty ZIP Code; however, this is most likely explained by the fact that those who search for

longer are much more likely to change ZIP Codes at all. This is reflective of considerably shorter search times for households who "lease in place", or use their voucher to pay for the same housing unit where they resided when they received their voucher.⁶

4.2 Monthly patterns in voucher issuance and outcomes

Importantly, there is also variation in search outcomes across calendar months of the year. Search times in particular show strong patterns across different months. Searches are shortest at the beginning of the year, and then increase throughout the spring and summer; they dip slightly in the fall, and then remain elevated in November and December. The median search time for vouchers issued in January is 61 days, but the median household who is issued a voucher in November searches for 10 days longer. The patterns in search time throughout the year differ from normal seasonal trends in the rental housing market. In general, both rental market demand and rents tend to increase throughout the spring, peak in June, and then decline throughout the rest of the year (appendix figure E1).

Success rates, on the other hand, show much lower variation across the year, and fluctuate very minimally across months, from a low of 66.7% in March to a high of 68.2% in October. It's also worth noting that other key metrics within the voucher program do not show the same seasonal trends as search time. For instance, households are about equally likely to move ZIP Codes throughout the year, including households with children who may be more inclined to make such moves in the summer when children are in between grades (appendix figure D3).

4.3 Allowable search times across PHAs

A question that arises from the stylized facts presented thus far is how much of the variation in search time comes from differences in allowable search time, as opposed to differences across households within a certain set of constraints. Data gathered from administrative plans can show both how long PHAs allow households to search by right, and how flexible they can be with extensions.

⁶I do not observe leasing in place in the data, as I don't have the household's full address at the time of search voucher issuance. I use an indicator for moving ZIP Codes as a conservative estimate of moving housing units between issuance and lease-up.

Across the 217 PHAs with available administrative plans from 2016-2020, the majority of PHAs give the federal minimum 60 day initial search term. However, PHAs almost universally allow extensions, and many PHAs even give automatic extensions.⁷ More information about the prevalence of extensions and other characteristics of PHA search time policies can be found in Appendix B.

Importantly, however, initial search times are not always closely related to observed search time. PHAs that have a 60 day initial search term actually tend to have longer searches, on average, than PHAs with 90 or 120 day search times (table 1). PHAs with 90 day effective initial search terms have the shortest observed search times. This indicates that much of the difference in search time across PHAs is likely due to how lenient PHAs are with granting extensions. Since administrative plans are approved only once a year, differences in observed search times across the calendar year are not plausibly due to changes in allowable search time.

5 Empirical strategy

The effects of search time are difficult to identify, given potential endogeneity issues. Part of this endogeneity comes from unobserved characteristics of households who search for longer, which can also affect their immediate and longer term housing outcomes. For instance, a household with multiple children may take longer to complete a housing search because it may be more difficult to find a larger rental units with sufficient bedrooms to house the family. In this case, the longer search time could be a representation of this challenge, instead of the households taking longer to find a unit that is a better match. In this paper, I will distinguish between households who take a long time to complete their housing search, and households who will search for longer when offered additional time.

Another potential confounding factor is that some of the variation in search time across housing authorities can be explained by differing search term and extension policies. Without more complete information on search time policies (i.e., policies for each PHA in each year), it is difficult to account for differences in the amount of time that households are *given* to search. For that reason, I use PHA by year fixed effects, to investigate the effect of observed search time, holding allowable search

⁷The difference between automatic and discretionary extensions is that in the case of an automatic extensions, generally households must still request the extension, but their request will not be denied. In the case of a discretionary extension, a PHA may deny the request or require the household to meet certain conditions before granting additional search time.

time constant. The effects can therefore be interpreted as the effects of variation in search time within a PHA.

The basis for the empirical design is that the timing of search voucher issuance is random, but that there are elements of issuance timing that can affect search time and the search process in general; namely, the month in which the search voucher is issued. As detailed in section 4.2, there are strong patterns in search time throughout the calendar year, that cannot be explained by rental market seasonality alone. Searches are longest in the summer months, but peak again in November and December; this may be due to reduced PHA capacity to conduct housing quality inspections and approve units. This means that households can receive additional search time at certain times of the year. This additional search time is the result of differences in both PHA practices and capacity at different times of the year.

This design relies on the assumption that timing of voucher issuance is random. Several facts support this assumption. First, most households wait for many years to receive their voucher; the average household waits around 2.5 years, and some households wait significantly longer, upwards of 10 years. Additionally, waitlists are frequently closed, so in practice households all enter the waitlist on the same day. This means that participants have little to no ability to manipulate the timing of their voucher issuance.

While PHAs have different prioritization strategies, these do not necessarily align with the calendar year. For instance, many PHAs prioritize households experiencing homelessness; however, they do not necessarily start a cycle of voucher issuance every January. PHAs may work through their waitlists with certain strategies, but the timing of waitlist opening and closing varies widely across the country, meaning the households are issued vouchers throughout the month and across months. Appendix Figure D1 shows that vouchers are issued across the months of the year.

A balance test of household and origin neighborhood characteristics confirms that issuance does appear to be random across months of the year (appendix figure D2). There are only slight deviations in the characteristics of households who are receiving vouchers from month to month, and no apparent patterns based on the calendar year.

I first estimate the effect of month of issuance on the probability of using a voucher, and the length of time that it takes for successful households to lease up. The equation is as follows:

$$Y_{itm} = \beta \sum_{j=1}^{12} \mathbb{I}(m - j = 0) + \gamma X'_{itm} + PHA_{itm} \times Year_t + \varepsilon \quad (1)$$

where j is an indicator for each month of the year, m is the month of issuance for household i in year t , X' is a vector of individual covariates including race, ethnicity, age, household composition, disability status, and homeless status. Models include PHA by year fixed effects, which means that PHA policies around allowable search times are held constant. These models are estimated using successful voucher use as the dependent variable, and with the log of search time as the dependent variable for only searches that are successful.

A second question is how allowable search time affects other outcomes for households who are successful in their search. If there is no effect of issuance timing on success rates, there is no selection into the sample of successful searches based on timing, and I can then use the month of issuance timing as an instrument for search time. As described in section 4.2 above, there is considerable variation in search time across months of issuance. This variation is not just a function of seasonal patterns in market tightness, but also reflects idiosyncratic differences in the voucher search process, and differences in how PHAs handle extensions throughout the year.

These models take the following two-stage least squares form:

$$Y_{itm} = \beta \log(\widehat{Searchtime}) + \gamma X'_{itm} + PHA_{itm} \times Year_t + \varepsilon \quad (2)$$

where the first stage is given by equation 1, using the log of search time as the dependent variable.

In addition to the assumptions detailed above, the standard IV assumptions apply. Figure 5 show that search time varies across days within a month and months within a year, which serves as the first stage for the IV models.

The exclusion restriction is also satisfied. The month of search voucher issuance is unlikely to be correlated with the outcomes of interest (neighborhood poverty rates and housing turnover), except through the effect on search time. This assumption is supported by the fact that issuance of search vouchers throughout the year is not associated with any characteristics of households with different propensities to move ZIP Codes, or different trends in housing turnover. The rate of moving ZIP Codes is also quite constant across months (appendix figure D3), both for all households and for

households with children. Lastly, households over 65 do not see the same pattern of search time. These households have a high propensity to lease homes in their origin ZIP Code, generally due to leasing in place; the fact that this group shows different patterns in search time throughout the year provides evidence against the claim that longer search times are due to delays in inspections or other seasonal patterns in the rental market.

One benefit of using an instrumental variables approach is that the local average treatment effect (LATE) is a useful tool for thinking about how search time can be used as a tool to improve programmatic outcomes. The LATEs in this study represent the effect of additional search time for households who will "take up" extra search time when their issuance timing allows; therefore, results are informative on how search time policy changes can affect match quality.

6 Results

6.1 Effect of issuance timing

The timing of voucher issuance may affect both the probability that a voucher recipient will be successful in using their voucher, and the amount of time that they search when they are successful. In Figure 4, we see that there the probability of successful voucher use within one year of issuance is relatively constant across months of the year, with the exception of the month of January, where the probability is slightly higher.⁸ Success rates are around 0.5-1.5 percentage points higher in January, relative to other months of the year.

However, there are strong effects on the amount of time that a household searches. After controlling for individual characteristics and PHA search policies, we still see that searches are long in the summer months, decline in September and October, and then rise again in November and December (Figure 5).

Given the very modest effect on successful voucher use, there is little concern about selection into the sample based on month of issuance; in other words, the month of voucher issuance has only a modest effect on the changes of making any match at all.⁹ The following analysis use the month

⁸Results are extremely similar when using a measure of lease-up success with no time window restriction; however, using a 1-year window ensures that results are comparable across years.

⁹As a robustness check, I show that results hold when excluding January altogether, and leveraging variation in search time across other months of the year.

of issuance as an instrument for search time, to test the effect of longer searches on neighborhood outcomes, housing turnover, and additional outcomes such as housing costs and unit size.

6.2 Effects of search time on neighborhood outcomes

Longer searches may permit households to reach lower poverty neighborhoods, a core but sometimes unfulfilled goal of the voucher program. In Table 4, OLS and IV regressions are shown for three different dependent variables: an indicator for moving ZIP Codes, an indicator for moving to a lower poverty ZIP Code, and an indicator for moving to a higher poverty ZIP Code. Focusing first on moving ZIP Codes at all, in the OLS model, we see that households who search for longer are more likely to move ZIP Codes. However, in the IV model, the coefficient is greatly reduced and insignificant. Intuitively, this is due to the fact that the instrumental variables approach identifies the treatment effect for "compliers", or households who will search for longer when their date of issuance allows them more time. A large portion of households with shorter search times do not move from their origin ZIP Code, and it is likely many of these household are remaining in their same home, which drives the result that longer searches associated with a lower chance of moving ZIP Codes. However, households who lease in place are unlikely to take up additional search time, so we see an insignificant causal effect of longer searches.

While there is no evidence that longer searches are inducing more moves, it does appear that longer searches may cause different moves among successful voucher recipients. The IV results on the probability of moving to a lower poverty ZIP Code indicate that a 50% increase in search time leads to a 2.5 percentage point increase in moving to a lower poverty ZIP Code, which is an increase of about 7 percent. This is a substantial effect, considering that often much higher touch interventions are required to help households move to a broader set of neighborhoods (Bergman et al., 2024; DeLuca et al., 2023).

In the last set of models, I look at the effect on moving to a higher poverty ZIP Code. Almost 30% of voucher holders actually move to a higher poverty ZIP Code than where they lived when they received their voucher. As seen in the summary statistics (table 3), longer searches are actually associated with a higher probability of moving to a higher poverty ZIP Code; this is confirmed in the OLS model. However, this is largely due to the fact that households who search for longer are more likely to move in general. When we use the IV strategy, the coefficient flips signs (though

insignificant in this specification), again reflecting the finding that longer searches help households reach lower poverty neighborhoods.

Given large difference in search times and search experiences across demographic groups and market conditions, I conduct heterogeneity tests to determine which groups are driving these effects. Black households and households with kids have positive and significant treatment effects for moving to a lower poverty neighborhoods (Figure 6). For older households, with a household head above 65, there is no effect of longer searches, which is consistent with the fact that older households are substantially less likely to move ZIP Codes when using their voucher, and may have less to gain from making an opportunity move.¹⁰ Hispanic households also show no effect of longer searches on moving to a higher opportunity neighborhoods.

Rental market dynamics appear to play a large role. Figure 6 shows the effects on moving to a lower poverty ZIP Code, stratified by the rent level of the county where most of the issuing PHA’s voucher holders live. Here, we see that effects on moving to a lower poverty ZIP Code are entirely driven by households in higher rent markets, and that the effect size in high rent markets is quite significant. The coefficient indicates that a 50% increase in search time translates into a 7.6 percentage point increase in the probability of moving to a lower poverty ZIP Code, or a 20% increase off a base rate of 36.7%. Theoretically, it makes sense that in tighter rental markets, additional search time may be needed to make an upwardly mobile move.

6.3 Effects of search time on housing turnover

Longer searches are associated with a lower chance of exiting the program or moving housing units within 1 to 2 years of entering the program (table 3). This result can be driven by households finding housing that better fits their needs and goals when they search for longer. In Table 5, I show OLS and IV regressions of search time on exiting the program within 1 or 2 years. Households are also more likely to stay in their same housing unit for longer when they search for longer. A 50% increase in search time causes a 3.5 percentage point decline in moving units within the following two years. This is a large decline as on average, around 14% of new voucher recipients will move within 2 years of their initial lease-up; this equates to a 20 percent drop in the probability of

¹⁰This is also consistent with findings from Ellen et al. (2024a), which finds that older households do not take-up a programmatic change that made vouchers more generous in higher opportunity areas.

moving. This result indicates that longer searches can benefit both voucher recipients and PHAs; households enjoy more stability and lower moving costs, and housing authorities may see reduced costs as moves involve conducting additional inspections and staff resources.

Looking at program exits, in the OLS model there is no significant effect; this differs from the raw trends shown in Table 3 due to controlling for differences across individuals and PHAs, where average search times can differ substantially. However, the IV regressions show negative treatment effect on exiting the program within 2 years, that is significant at the 10% level. Specifically, a 50% increase in search time leads to a 2.3 percentage point decline in exiting the program within 2 years. This is a substantial decline off of a base rate of 18%. The average voucher holder remains in the program for 10 years; exits within just 2 years of entering may be indicative of dissatisfaction with their housing unit, or an inability to find another suitable unit.

In terms of heterogeneity, effects on housing unit turnover are predominately driven by Black households (Figure 7). This is an encouraging results for several reasons. First, Black households search for significantly longer than White households, even when controlling for other factors (appendix figure C1), and second, Black households have elevated rates of moving shortly after entering the program. Effects on moving housing units are also driven predominately by younger households, but are significant for both households with kids and households without children. And while effects are slightly larger in tighter rental markets, effect sizes are not significantly different.

Heterogeneity in the effects of search time on exiting the program within two years show a different pattern. Treatment effects on program exits are driven by white households, and are also much stronger in higher rent markets. This may be due to the fact that White households are actually significantly more likely to exit the program within two years of entry (25% of white households exit the program within two years, compared to just 16% of Black households and 15% of Hispanic households). Additional search time may aid white households in finding housing units that avoid quick turnover within the program.

6.4 Additional outcomes

In addition to neighborhood mobility and match quality, there are other margins that longer searches may affect. One of these is housing costs. While low-income renters with a traditional budget constraint may try to optimize on price when given more time to search, voucher holders

do not experience any added price increases up to the payment standard. In fact, given the results on moving to lower poverty neighborhoods, voucher holders may actually lease up into higher rent units when they search for longer. Table F1 confirms this hypothesis, showing that longer searches lead to higher gross rents, which translates into higher housing authority portions of the rent. However, there is no increase in the share of renters who have a rent burden over 30%, which indicates that voucher recipients that search longer rent more expensive units, but on average do not exceed payment standards, above which they would be obligated to pay the difference between the gross rent and the payments standard out-of-pocket.

Another margin that voucher recipients may optimize on is the size of their rental unit. When a household is issued a voucher, they qualify for a certain bedroom size based on the number, age, and sex of household members.¹¹ At the time of lease-up, households are generally not permitted to be "under-housed", or renting a unit with fewer bedrooms than they are qualified for, unless they receive special approval. However, households may be "over-housed", or rent a unit with more bedrooms than they are qualified for, given the unit falls within the payment standard for the smaller bedroom size that they qualified for.

Households who search for longer may attempt to increase the size of their housing unit; in this case, we would expect longer searches to cause a drop in being under-housed, and an increase in being over-housed. In Table F2, we see that there is no significant effect on either outcome, though the signs of the coefficients are in the expected direction. Table F3 shows the same outcomes, stratified by market conditions. Here, there is marginally significant increase in being over-housed, for those in low rent markets. This suggests that longer searches may allow voucher recipients to rent larger units, but only where rents are low enough that larger units are still affordable through the voucher program.

¹¹PHAs have some flexibility in setting bedroom size requirements, as long as they are not less generous than federal standards.

7 Robustness

7.1 Additional IV specification: day of voucher issuance

In addition to variation in search time across months of the year, there can also be variation within a month, based on the day that a voucher is issued. This is because vouchers can be issued on any day of the month (appendix figure G1, left panel), but households generally only begin their leases on the first of the month. This means searches can essentially be different lengths based on when the search starts. Searches at the end of the month tend to be longer, on average, as households are likely to roll over and begin their lease a month later (appendix figure G1, right panel).

While the day of the month does induce changes in search time, on it's own, the variation is not strong enough to use as an instrument in the empirical analysis. On average, searches at the end of the month are only about a day longer than those in the first three weeks of the month (appendix table G1). However, when used in combination with the month of issuance, the day of issuance can create significant variation in search times, and provides an alternative specification that can confirm the core results presented above.

Leveraging this additional source of variation, I replicate my core results with three additional IV specifications. The first uses the week of the year of voucher issuance as an instrument (from 1 to 52). This essentially combines the variation we see from the month of year and the timing within the month. I also use two different versions of interacted instruments; one indicator for the week within the month (from 1 to 4, with 4 being the last 7 days of the month, 3 the 7 days prior, 2 the 7 days prior to that, and the remaining days grouped together as the beginning of the month), interacted with the month of the year, and an indicator for the last 7 days of the month, also interacted with the month of the year. Appendix figures G2 and G3 use the final specification and show that first, there is again no effect on successful voucher use (no selection into the sample based on the instrument), and second that the interaction of these two variables does in fact create meaningful differences in search time.

Appendix tables G2 and G3 show that core results for housing turnover and neighborhood outcomes are generally very consistent across these additional empirical specifications.

7.2 Tests for weak instruments

I also conduct a series of tests to ensure that results using my primary IV specification, the month of voucher issuance, are robust under weak instruments. Anderson-Rubin tests on the three core outcomes (moving to a lower poverty ZIP Code, moving within 2 years, and exiting within 2 years) all yield p-values under 5%, indicating that the treatment effects are robust under weak instruments.

7.3 Sample exclusions

Finally, I confirm that results are robust to different sample restrictions. Appendix tables [H1](#), [H2](#), and [H3](#) replicate core results for moving within two years, exiting within 2 years, and moving to a lower poverty ZIP Code. First, results are qualitatively similar when dropping 2020, a year that may have been impacted by the COVID-19 pandemic, and households that could have been impacted by the near-universal and significant rise in rents and rental market tightness that followed. I also confirm that results hold when dropping households that receive assistance through a special purpose voucher (SPV); some types of SPVs come with additional search support or longer initial search terms that could influence either set of core outcomes. I also drop both homeless and disabled households; households who are experiencing homelessness at the time may receive extra assistance in the search process, and almost all PHAs offer extended time as a reasonable accommodation to households with disabilities. I also include one specification that excludes all the aforementioned categories. Across all samples, coefficients are qualitatively similar.

8 Conclusion

The voucher program is one of our most important tools for ensuring that low-income households can find safe and affordable housing. However, the strenuous private market housing search process can prohibit families from using their vouchers, and lead to inefficient housing matches when they do use their vouchers. These results provide evidence that lengthening search times may be an effective tool in promoting better housing matches among voucher recipients.

The second component of this project involves looking at another source of variation in search times: allowable search time. Original data collection is in progress that will allow for testing the

effect of different search term policies, both on success rates and on these same margins of match quality. This will provide further evidence on what PHAs stand to gain from modifying their program operations.

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Tables and figures

Figure 1: Success rates and search times, 2015-2021

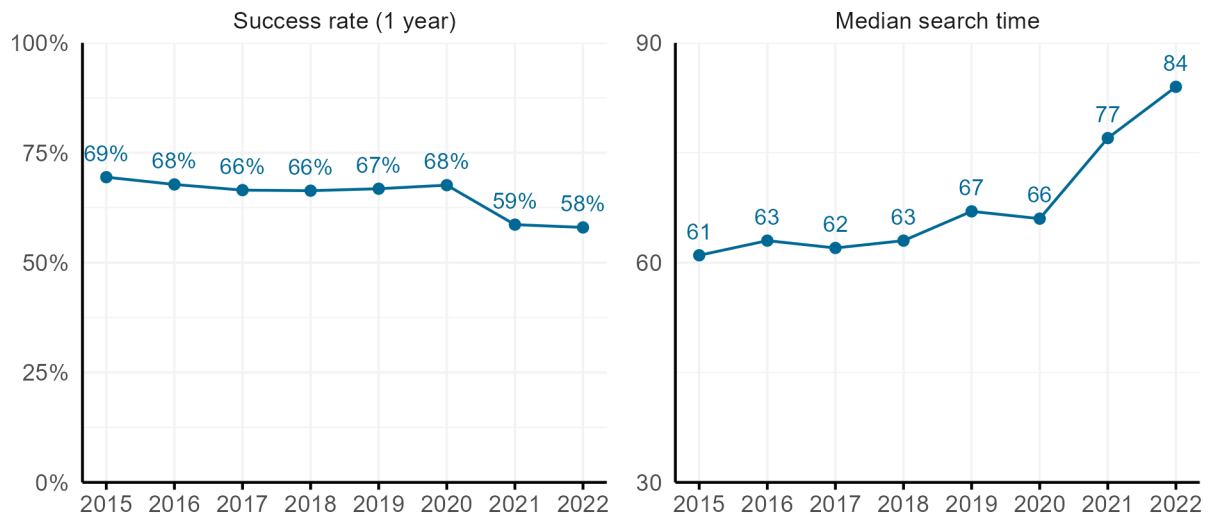


Figure 2: Distribution of success rates and search times, 2019-2020

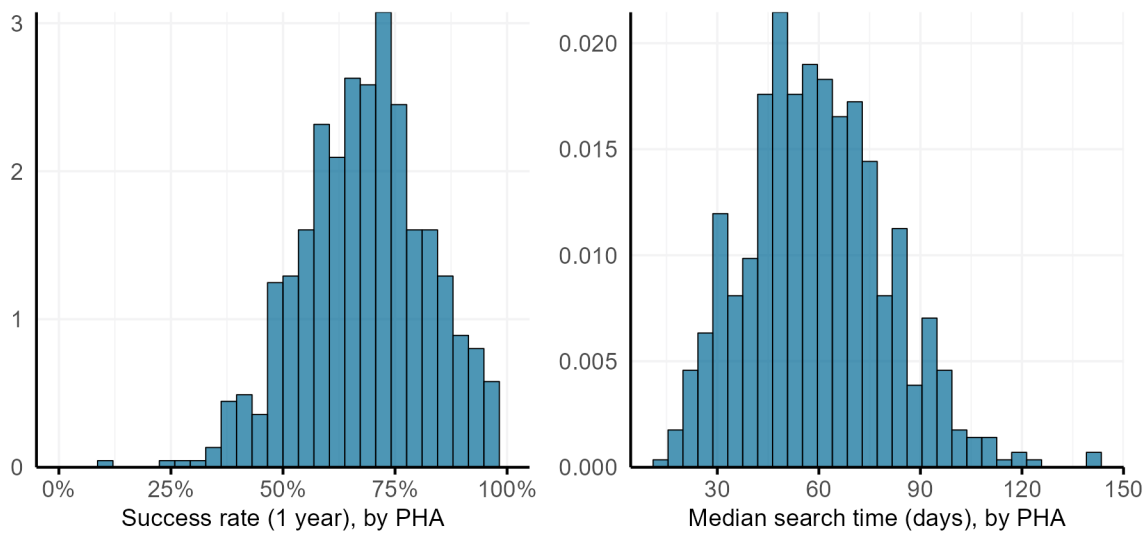


Figure 3: Raw, uncontrolled success rates and search times by month of voucher issuance

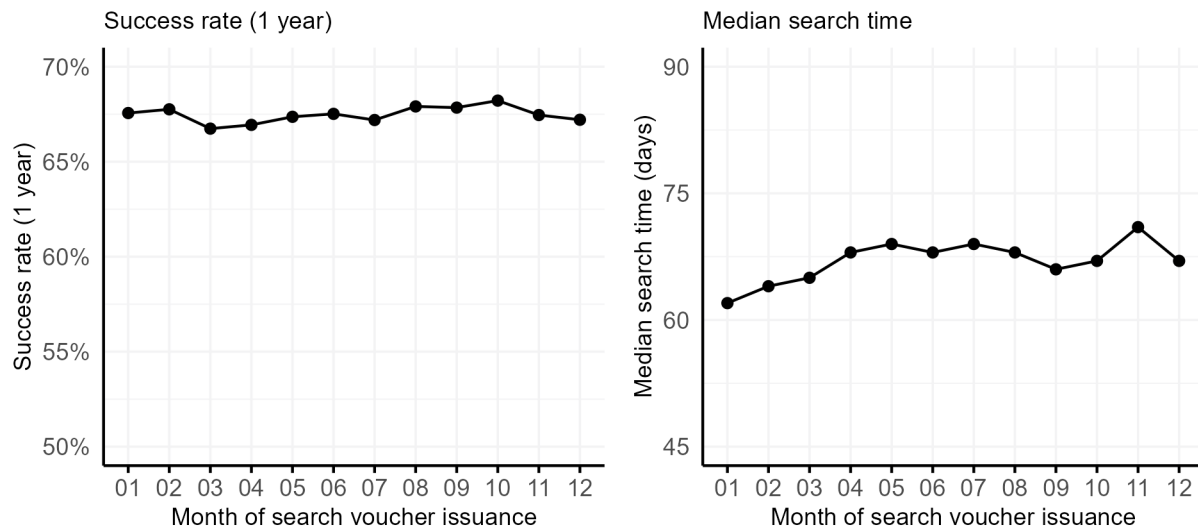


Table 1: Observed search time by effective initial search term

Effective initial voucher term	Number of PHAs	Number of successful searches	Median search time	Share of searches at least 15 days longer than effective term
60	100	24475	75	0.496
90	32	8382	68	0.247
120	29	6748	72	0.198

Note: Includes search events in the same effective year administrative plan collected. Search time is calculated for successful searches only.

Table 2: Sample summary statistics

	Observations	Share of observations	Success rate (1 year)	Median search time
Black head of household	307395	0.510	0.687	74
Hispanic head of household	85576	0.142	0.673	62
White head of household	183806	0.305	0.663	50
Other race head of household	26440	0.044	0.645	57
Female head of household	434820	0.721	0.682	66
Presence of kids	303404	0.503	0.690	68
Age 65+	60210	0.100	0.690	46
Disability indicator	185363	0.307	0.674	60
All observations	920593		0.648	67

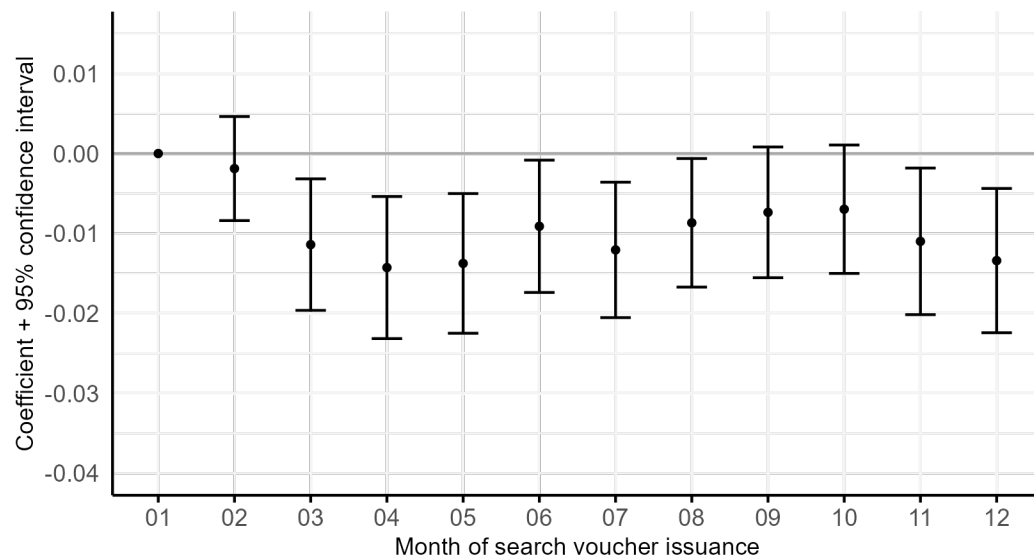
Note: Includes successful searches from 2015-2020.

Table 3: Raw, uncontrolled outcome variables, by search time

Search time	Housing turnover				Neighborhood outcomes		
	Exit within 1 year	Exit within 2 years	Move within 1 year	Move within 2 years	Move ZIP Codes	Move to lower poverty ZIP Code	Move to higher poverty ZIP Code
29 days or less	0.103	0.227	0.043	0.134	0.486	0.253	0.222
30-59 days	0.085	0.196	0.042	0.142	0.606	0.329	0.270
60-89 days	0.077	0.182	0.038	0.141	0.673	0.365	0.302
90-119 days	0.069	0.171	0.037	0.138	0.700	0.382	0.313
120+ days	0.060	0.153	0.034	0.126	0.731	0.392	0.334
All successful searches	0.079	0.186	0.039	0.137	0.639	0.344	0.288

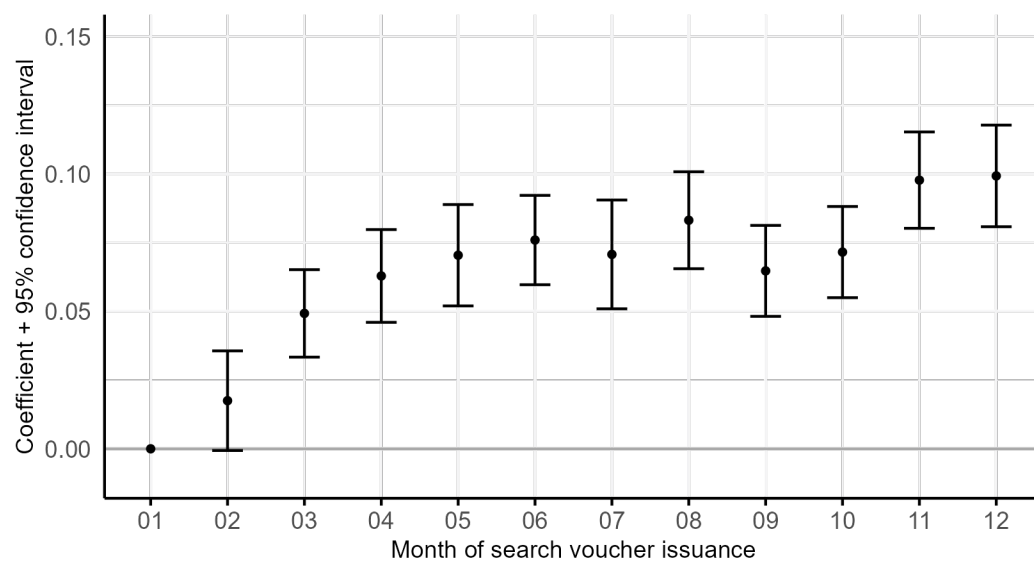
Note: Includes successful searches from 2015-2020. Moves are defined as a change of building.

Figure 4: Effect of month of issuance on leasing success



Note: The dependent variable is an indicator for the household successfully using their voucher within one year of issuance. Model includes individual controls (race, ethnicity, age, household composition, homeless status, citizenship status, disability status) and PHA by year fixed effects. Standard errors are clustered by PHA.

Figure 5: Effect of month of year on search time



Note: The dependent variable is the natural log of search time, for households who are successful in using their voucher. Model includes individual controls (race, ethnicity, age, household composition, homeless status, disability status) and PHA by year fixed effects. Standard errors are clustered by PHA.

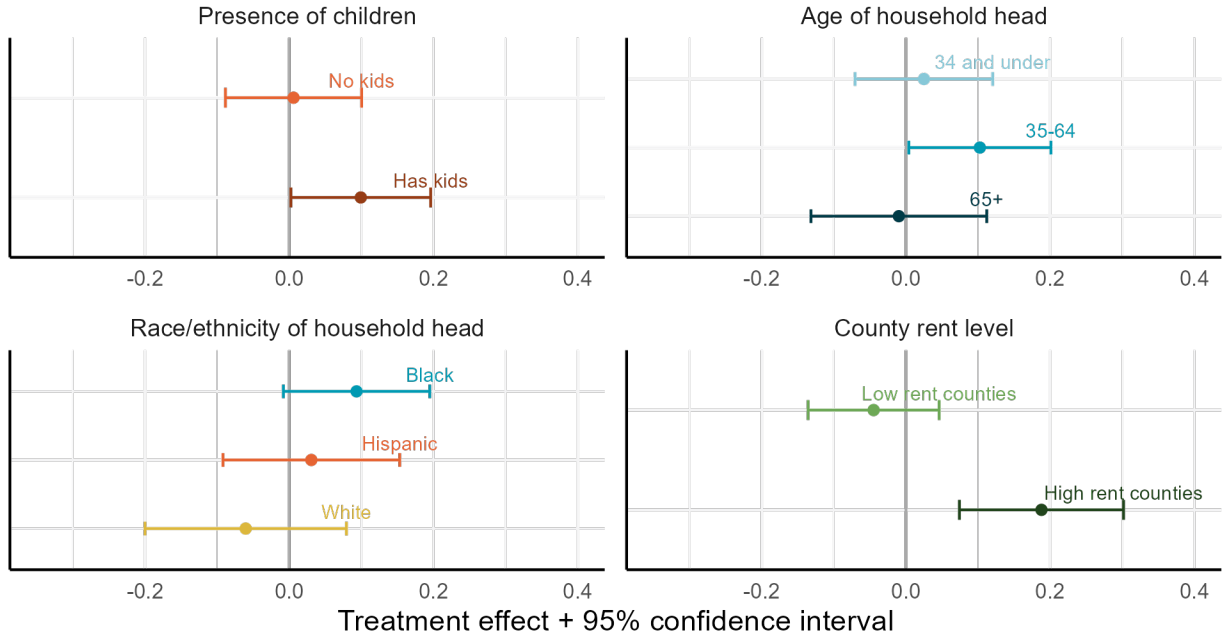
Table 4: OLS and IV estimates on neighborhood outcomes

	Move ZIP Codes		Move to a lower poverty ZIP Code		Move to a higher poverty ZIP Code	
	OLS	IV	OLS	IV	OLS	IV
log(Search time)	0.068*** (0.002)		0.032*** (0.002)		0.037*** (0.002)	
log(Search time), fitted		0.025 (0.036)		0.062+ (0.038)		-0.043 (0.032)
Num.Obs.	431 991	431 991	423 381	423 381	423 381	423 381
PHA by Year fixed effects	X	X	X	X	X	X
Individual controls	X	X	X	X	X	X
First stage F stat		45		44.8		44.8

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Month of issuance and an indicator for the last week of the month are used as instruments. Standard errors are clustered by PHA.

Figure 6: Heterogeneity in effect on moving to lower poverty neighborhood



Note: The dependent variable is an indicator for the destination poverty rate being lower than the origin poverty rate. Month of issuance is used as an instrument for search time. Model includes individual controls and year by PHA fixed effects. Standard errors are clustered by PHA.

Table 5: OLS and IV estimates on housing turnover

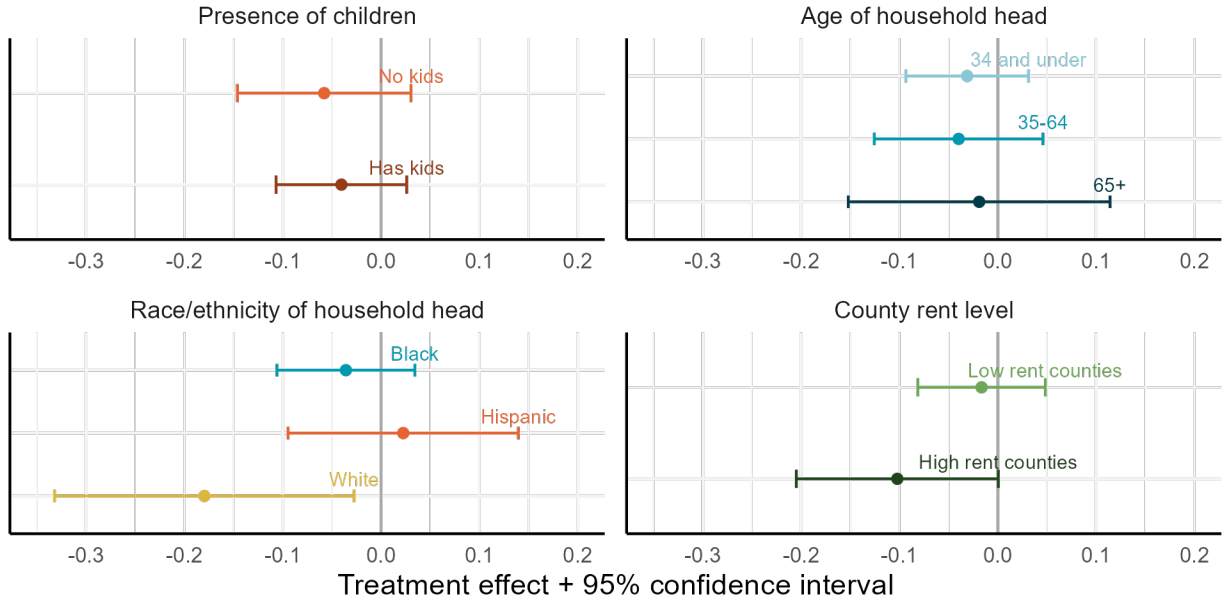
	Move within 1 year		Move within 2 years		Exit within 1 year		Exit within 2 years	
	OLS	IV	OLS	IV	OLS	IV	OLS	IV
log(Search time)	-0.003*** (0.000)		-0.006*** (0.001)		0.000 (0.001)		-0.001 (0.001)	
log(Search time), fitted		-0.015 (0.013)		-0.086*** (0.025)		-0.018 (0.020)		-0.057+ (0.031)
Num.Obs.	431 991	431 991	431 991	431 991	431 991	431 991	431 991	431 991
PHA by Year fixed effects	X	X	X	X	X	X	X	X
Individual controls	X	X	X	X	X	X	X	X
First stage F stat		45		45		45		45

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

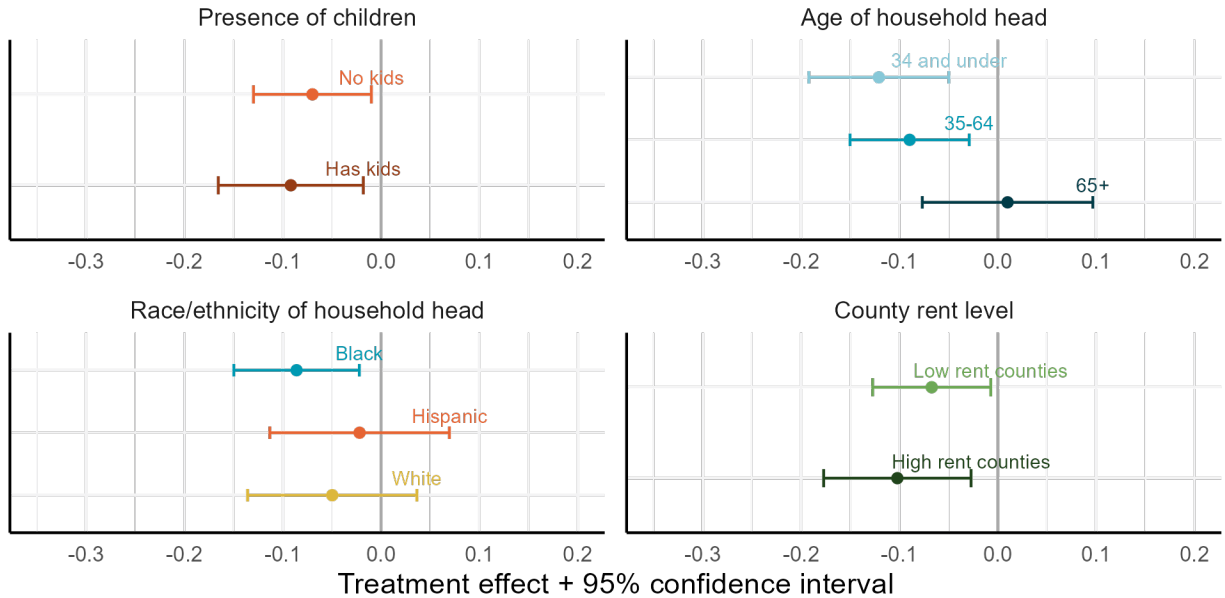
Note: Month of issuance and an indicator for the last week of the month are used as instruments. Standard errors are clustered by PHA.

Figure 7: Heterogeneity in the effect of search time on housing turnover

Panel A: Heterogeneity in effect on exiting the program within 2 years



Panel B: Heterogeneity in effect on moving within 2 years



Note: The dependent variable in panel A is an indicator for the household exiting the program within 2 years of entering. The dependent variable in panel B is an indicator for the household moving housing units within 2 years of entering. Month of issuance is used as an instrument for search time. Model includes individual controls and year by PHA fixed effects. Standard errors are clustered by PHA.

Appendix

Appendix A: Administrative data on voucher use: data quality standards

This paper uses administrative data from HUD to derive outcomes on success rates and search time. The process to clean these data was extensive; it is described in detail in Ellen et al. (2023). In order to ensure accuracy, I employ data quality standards to ensure that only PHAs with accurate and complete issuance data are included in the sample. In general, data quality standards ensure that I exclude PHAs that do not accurately report their issuance data; these PHAs are likely to have success rates that are biased upwards, as they could be under-reporting failed searches.

There are 1,213 PHAs that meet data quality standards across the seven year panel period from 2015-2022. These PHAs collectively serve around 71% of voucher holders. I then further restrict the sample to PHAs that are in the top 300 largest CBSAs, by number of voucher holders. In total, the top 300 CBSAs cover over 1,200 PHAs, or 77% of voucher holders. Legacy Moving-To-Work (MTW) PHAs are also excluded; these PHAs have different reporting requirements, and do not report origin ZIP Code, which is used to derive some of the key outcomes.

Combining these restrictions results in a final sample of 722 PHAs, which collectively cover 55% of voucher holders (using the population of voucher holders in 2019). Table A1 shows that searches that originate at these PHAs look relatively similar to all PHAs, and all PHAs that meet data quality standards.

Table A1: Balance test for inclusion in sample

Variable	All PHAs	PHAs that meet data quality standards	PHAs in top 300 CBSAs	PHAs in final sample
Black head of household	0.426	0.444	0.487	0.498
Hispanic head of household	0.145	0.140	0.156	0.151
White head of household	0.384	0.371	0.309	0.304
Female head of household	0.726	0.724	0.710	0.711
Presence of children	0.477	0.476	0.476	0.477
Age 65+	0.113	0.110	0.118	0.114
Disabled head of household	0.320	0.322	0.323	0.324
Noncitizen	0.026	0.025	0.032	0.031
Homeless indicator	0.105	0.106	0.125	0.125
Observations	2,207,503	1,712,372	1,498,546	1,017,694

Appendix B: Collecting data on allowable search times

While PHAs are mandated to give voucher recipients 60 days to search when they receive a voucher, there is no maximum amount of time, and PHAs have broad discretion to set initial search time and extension policies. There is, however, no centralized source of information about these policies. From HUD’s Housing Choice Voucher guidebook:

PHAs must give all voucher holders a minimum of 60 days search time. There is, however, no limit on how much longer a PHA may allow a family to search for housing. There are differing opinions on what length of search time is effective in promoting family success in leasing.

HUD also notes that PHAs that use longer search times should maintain data on searches so that they can monitor if extended search times could improve success rates, and to have knowledge of their potential financial liabilities. To my knowledge, this is the first paper that provides descriptive evidence on search time policies.

Each year, PHAs submit administrative plans, where they document a comprehensive set of policies that guide the day-to-day operations of the programs. PHAs are required to document their initial search time, or how many days the voucher is issued for; their policies around extensions, which can include automatic and discretionary extensions; their policies on voucher suspension; and their policies around expiration, or what happens when a household fails to use their voucher in the allotted time given.

For this paper, I collect data on these policies by collecting administrative plans from several sources. First, Zhang and Johnson (2023) graciously provided over 1,200 administrative plans and other documents from PHAs, collected between 2016 and 2020. To my knowledge, this is one of the only sources of historical plans, as most PHAs post only their more recent administrative plans on their websites. Of the documents provided, 217 PHAs are in my empirical sample, and meet data quality standards for the time period. For these PHAs, I can observe their search time policies, and see how these policies are reflected for both failed and successful searches.

Supplementary plans were collected directly from PHA websites, and from a FOIA request to the HUD Boston Regional office. Combined, these sources added an additional 55 administrative plans to the sample obtained from Zhang and Johnson (2023).

In Figure B1 below, I show the distribution of the effective year of the analyzed plans. In the analysis that follow, I use only plans that were in effect prior to 2021, and I exclude plans with an unknown year. In Figure B2, I show that most PHAs tend to adopt the minimum 60 day initial search window. While I do not have many PHAs that are represented across multiple years of the panel, it does appear that the share of PHAs with a 60-day initial search term did decrease in 2019 and 2020, which would be consistent with increasing search times.

Discretionary extensions are almost universal, and many PHAs offer automatic extensions as well (figure B1). Almost all PHAs will suspend the voucher term (known as “tolling”) while the PHA completes an inspection; this was made mandatory by HUD in 2015. Additionally, almost all

Figure B1: Effective year of plans analyzed

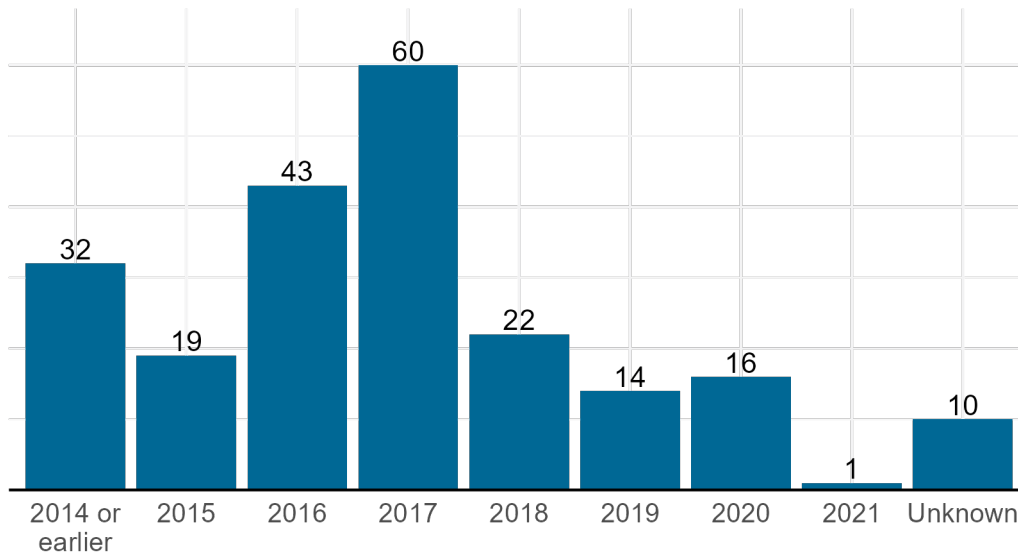
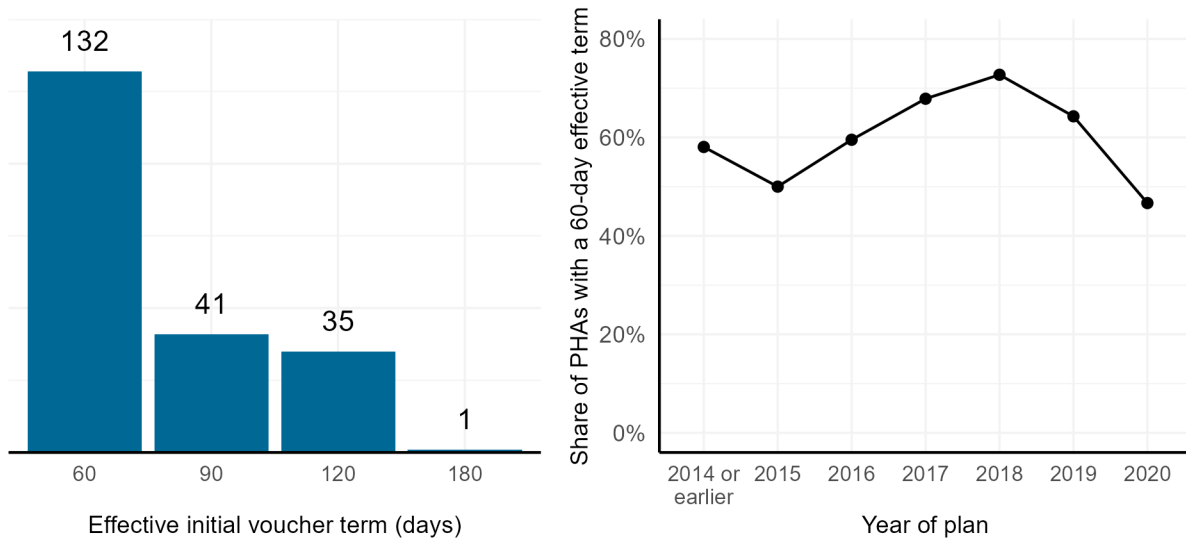


Figure B2: Effective initial voucher term



PHAs over additional time as a reasonable accommodation to voucher recipients with disabilities. PHAs will also often provide extra search time at issuance to households receiving certain types of special purpose vouchers.

Table B1: Share of PHAs that allow for automatic extensions, discretionary extensions, and tolling

Year of plan	Number of PHAs	Share that allow automatic extensions	Share that allow discretionary extensions	Share that suspend voucher term during inspections
2014 or earlier	32	0.156	0.875	0.800
2015	19	0.368	0.895	0.765
2016	43	0.256	0.977	0.923
2017	60	0.117	0.967	0.912
2018	22	0.227	0.955	0.850
2019	14	0.214	1.000	0.929
2020	16	0.375	1.000	1.000

Appendix C: Stylized facts about search time

Figure C1: Median search time by race/ethnicity and age of household head

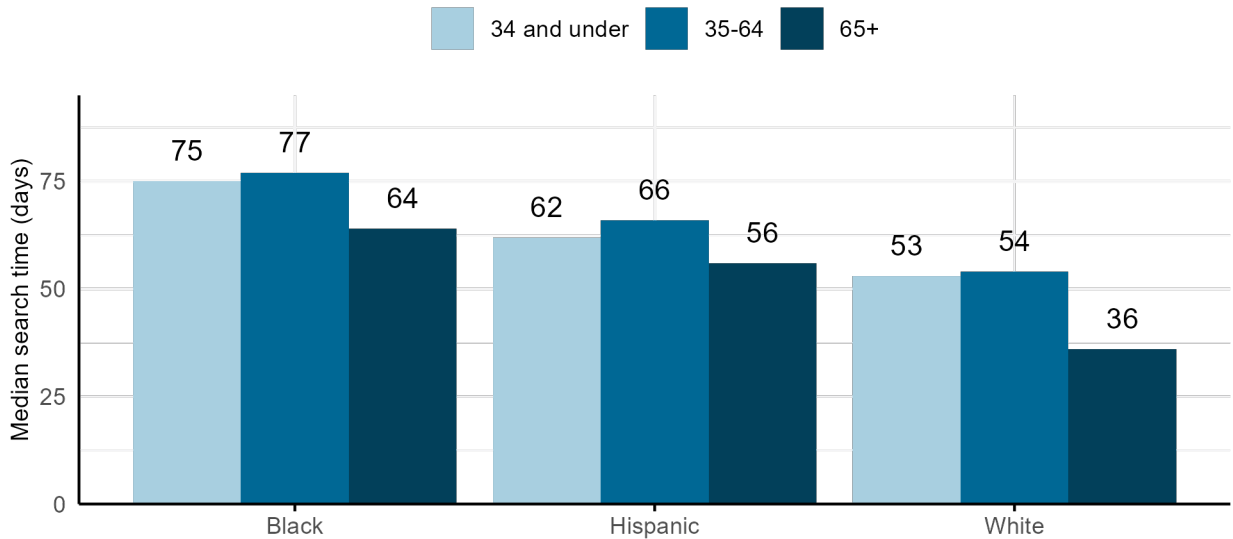
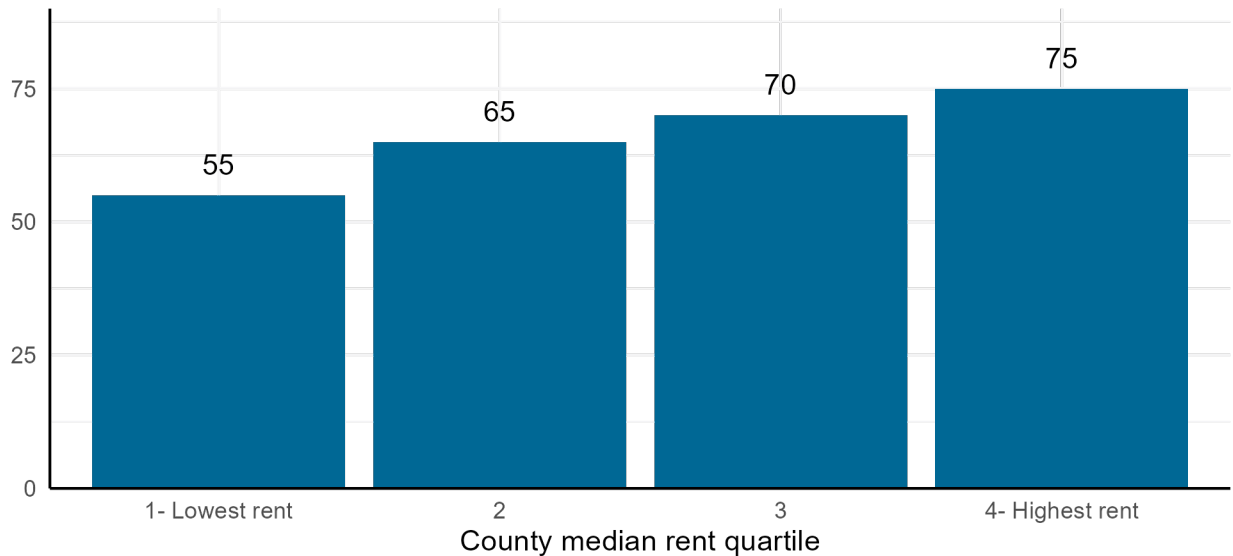
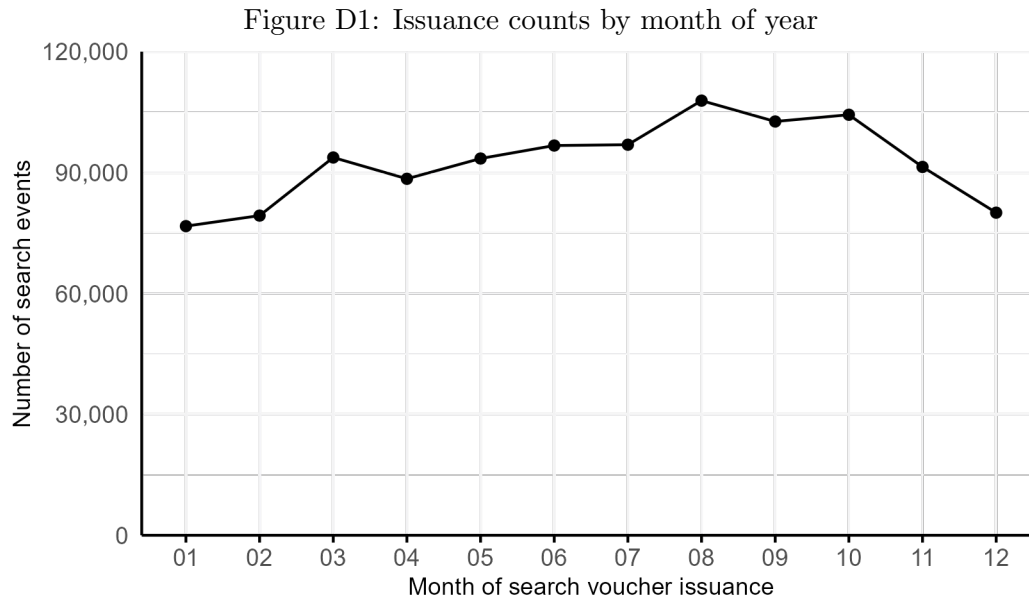


Figure C2: Median search time by county median rent quartile

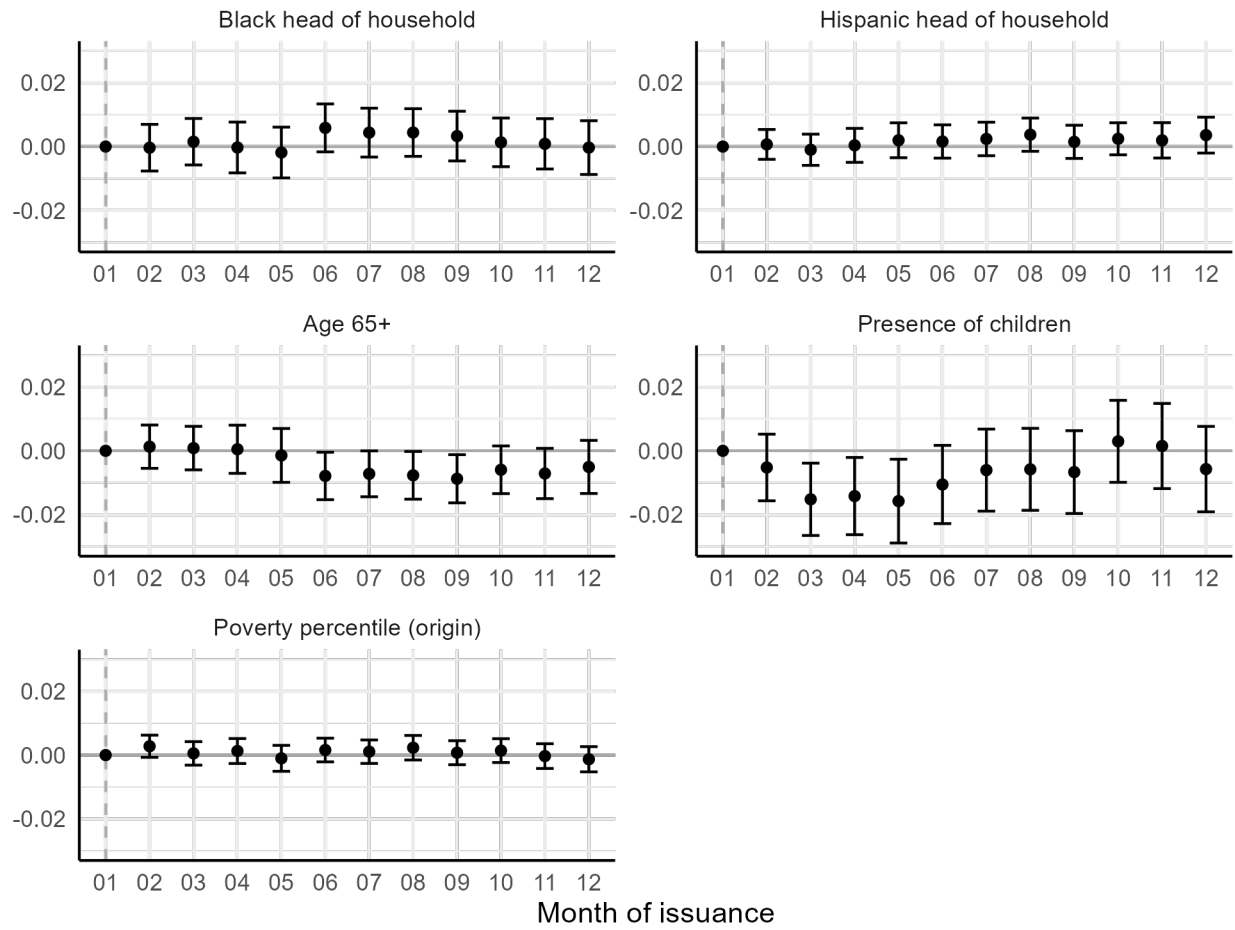


Appendix D: Voucher issuances and outcomes across months of the year



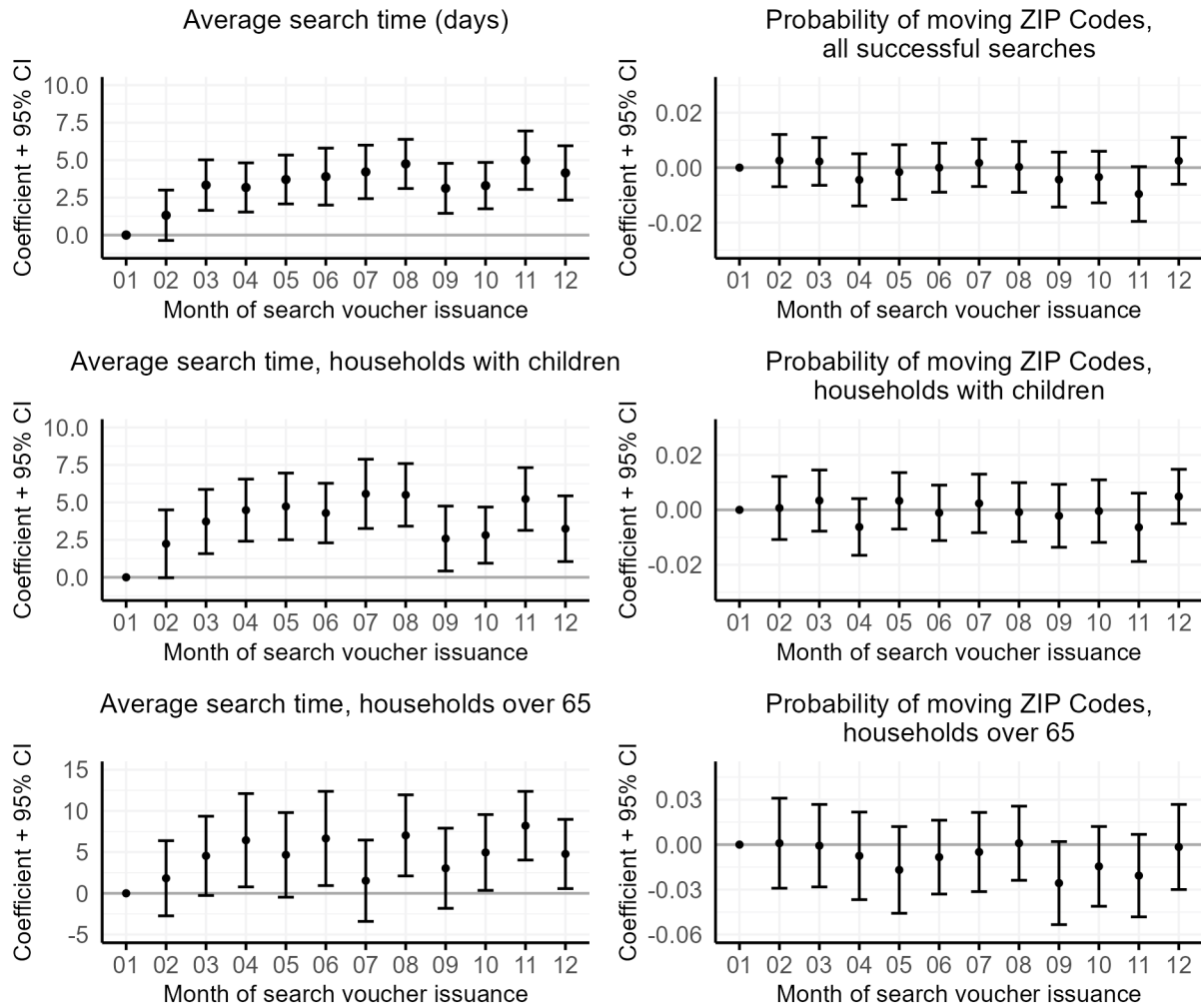
Note: The graph shows the total number of search vouchers issued each month from 2015-2020.

Figure D2: Balance test by month of issuance



Note: Each panel shows the coefficients from a model with a household characteristic as the dependent variable. Models include year x PHA fixed effects. Standard errors are clustered by PHA.

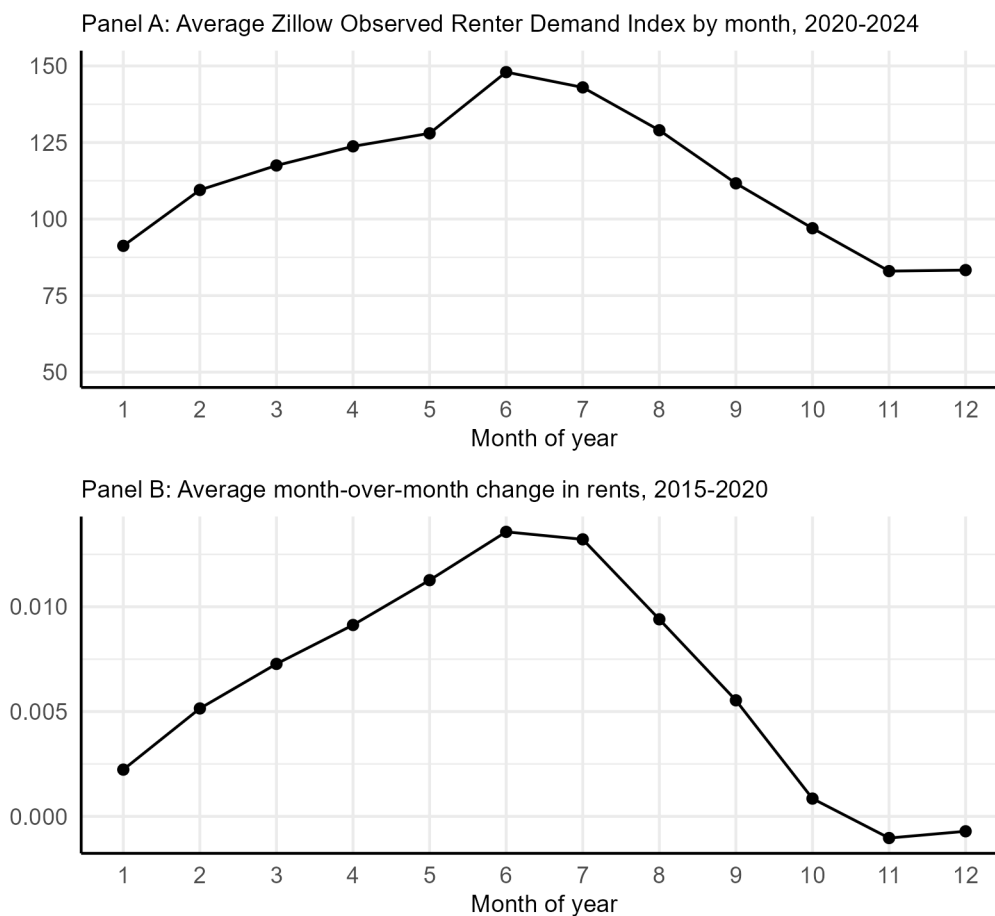
Figure D3: Trends in mobility outcomes by month of issuance



Note: All models include year x PHA fixed effects. Standard errors are clustered by PHA.

Appendix E: Seasonality in the broader rental housing market

Figure E1: Rental market indicators by calendar month



Note: Panel A averages the value of the Zillow Observed Renter Demand Index by calendar month of year, for all years that the index is available. Panel B uses the non-seasonally adjusted Zillow Observed Rent Index to obtain month-over-month change in rents; these values are then averaged across calendar months, using the years 2015-2020 to match the sample period.

Appendix F: Additional outcome variables

Table F1: Effect of search time on housing cost variables

	log(Gross rent)		Rent burden		Rent burden over 30%	
	OLS	IV	OLS	IV	OLS	IV
log(Search time)	0.019*** (0.001)		0.001*** (0.000)		0.014*** (0.002)	
log(Search time), fitted		0.358*** (0.045)		0.012 (0.009)		−0.004 (0.040)
Num.Obs.	431 962	431 962	383 430	383 430	383 430	383 430
PHA x Year fixed effects	X	X	X	X	X	X
Individual controls	X	X	X	X	X	X
Instrument		Month of year		Month of year		Month of year
First stage F stat		45.2		41.1		40.9

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Regressions of rent burden and an indicator variable for rent burden exceeding 30% exclude households with no reported income.

Table F2: Effect of search time on relative unit size

	Under-housed		Over-housed	
	OLS	IV	OLS	IV
log(Search time)	0.010*** (0.002)		0.000 (0.002)	
log(Search time), fitted		−0.012 (0.031)		0.032 (0.021)
Num.Obs.	432 615	432 615	432 615	432 615
PHA by Year fixed effects	X	X	X	X
Individual controls	X	X	X	X
Instrument		Month of year		Month of year
First stage F stat		59.9		59.9

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

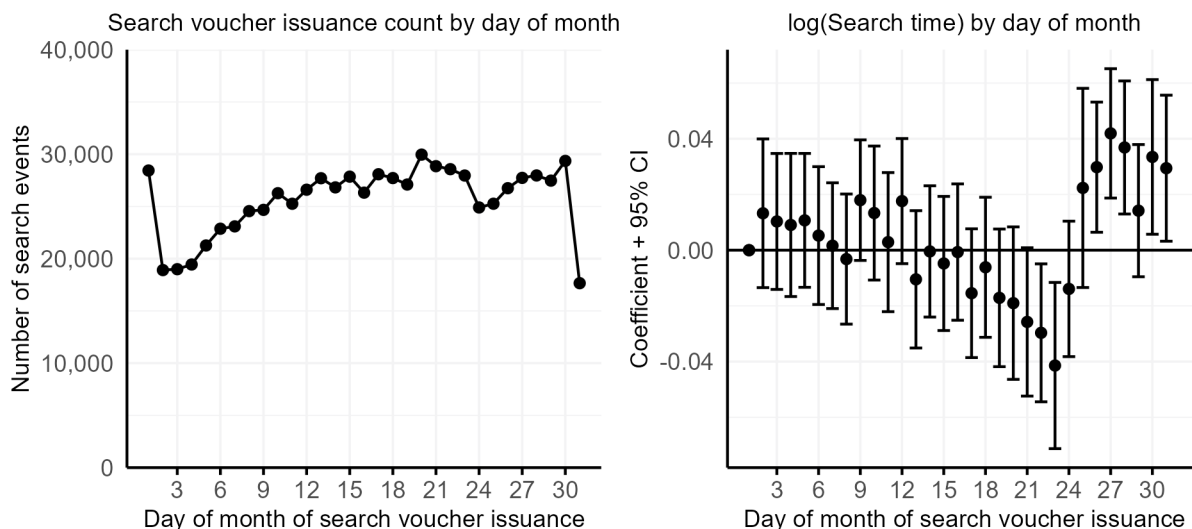
Table F3: Effect of search time on relative unit size, stratified by origin ZIP Code rent

	Under-housed		Over-housed	
	Low rent markets	High rent markets	Low rent markets	High rent markets
log(Search time), fitted	-0.004 (0.035)	-0.021 (0.051)	0.030+ (0.016)	0.042 (0.048)
Num.Obs.	236 600	196 015	236 600	196 015
PHA x Year fixed effects	X	X	X	X
Individual controls	X	X	X	X
First stage F stat	38.1	24.3	38.1	24.3
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001				

Appendix G: Alternative IV specification: using variation within months

As described in Section 7.1, there is some additional variation in search time that comes from the day of the month that a voucher is issued. In figure G1 below, I first show that vouchers can be issued throughout the month (left panel), and there is a mechanical relationship with search time throughout the days of the month (right panel). Searches tend to be shorter in the middle of the month, and longest at the end of the month.

Figure G1: Issuance count and differences in $\log(\text{Search time})$ by day of month of voucher issuance



However, when we translate this effect into days and compare within weeks of the month, the actual variation induced is very small. Searches in the last week of the month are just over a day longer, on average, than searches are other points in the month (table G1).

However, combining the month of issuance with the timing of issuance within a month can create more variation in search time. I combine these two sources variation in several different ways. First, I can interact the week of the month or an indicator for the last week of the month with the month of the year. Second, I can use the week of year directly as an instrument, where the value ranges from 1 for the first week of the year to 52 for the last.

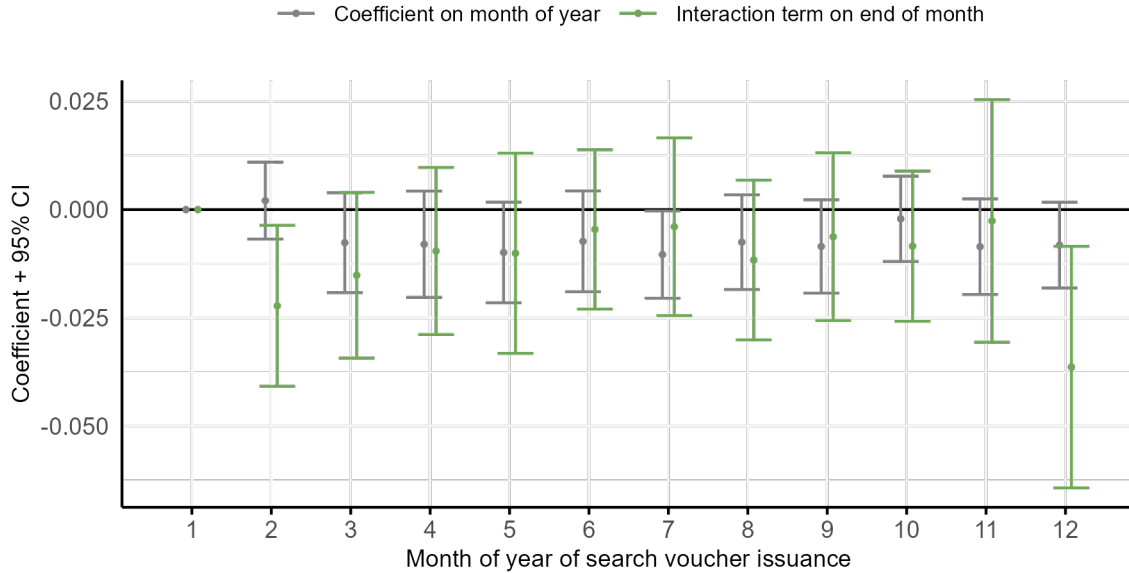
First, in Figure G2, I show that again, there is no substantial effect of the month of issuance or the interaction term between month of issuance and an indicator for the last week of the month on the ability to successfully use a voucher. In other words, receiving a search voucher at the end of the month in a high-search time month does not effect the ability of voucher recipients to use their vouchers; there is no selection into the sample based on timing within the month or the month of the year. Second, in Figure G3, I show that both the week of the year and the interaction term between the month of year and the high search time period at the end of the month create meaningful deviations in search time, though these deviations are not necessarily uniform across

Table G1: Effect of timing within month on search time

	Effect on search time (days)	
	Week of month, counting backwards	Indicator for last week of month
Third to last week	0.470 (0.292)	
Second to last week	-0.380 (0.355)	
Last week of month	1.151* (0.493)	
Indicator for last week of month		1.113** (0.389)
Num.Obs.	430 227	430 227
R2	0.164	0.164
R2 Adj.	0.157	0.156
PHA by Year fixed effects	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

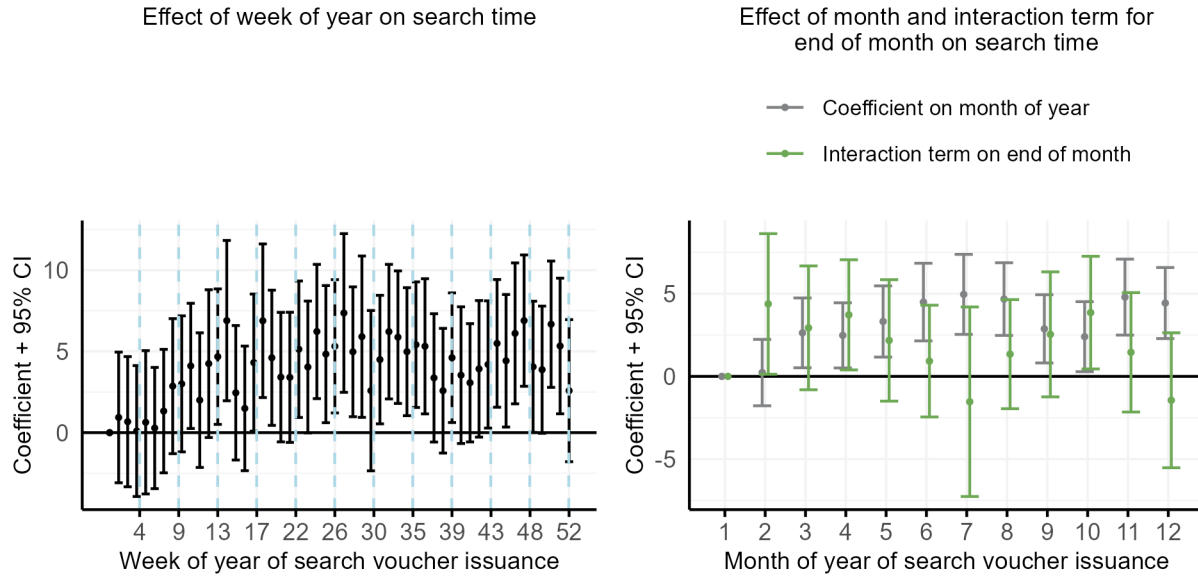
Figure G2: Effect of month of year and interaction term for the end of month on successful voucher use



months of the year.

In table G2, I show the effect of search time on moving to a lower poverty ZIP Code, using the primary specification with month of year as the instrument, as well the additional specifications described above. Results are generally quite consistent across models, with all specifications showing a both statistically and economically meaningful effect, though there is some variation in effect size. Interacting the end of the month with the month of the year carries the strongest first-stage

Figure G3: Effect of week of year and interaction term for the end of month on search time



F statistic. In table G3, I show that results on moving within 2 years are also robust to different instruments, though again with some variation in effect size.

Table G2: Effect of search time on moving to a lower poverty ZIP Code, using multiple instruments

	Dependent variable = Move to a lower poverty ZIP Code			
	Month of year	Week of year	Week of month x Month of year	End of month x Month of year
log(Search time), fitted	0.062+ (0.038)	0.050+ (0.028)	0.046+ (0.027)	0.057+ (0.031)
Num.Obs.	423 381	423 381	423 381	423 381
R2	0.090	0.092	0.092	0.091
R2 Adj.	0.082	0.083	0.083	0.082
Year x PHA fixed effects	X	X	X	X
Individual controls	X	X	X	X
Instrument	Month of year	Week of year	Week of month x Month of year	End of month x Month of year
First stage F stat	44.8	14.1	18	54.4

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table G3: Effect of search time on moving within 2 years, using multiple instruments

	Dependent variable = Move within 2 years			
	Month of year	Week of year	Week of month x Month of year	End of month x Month of year
log(Search time), fitted	−0.086*** (0.025)	−0.068*** (0.020)	−0.051** (0.018)	−0.059** (0.021)
Num.Obs.	431 991	431 991	431 991	431 991
R2	0.012	0.023	0.030	0.027
R2 Adj.	0.003	0.013	0.021	0.018
Year x PHA fixed effects	X	X	X	X
Individual controls	X	X	X	X
Instrument	Month of year	Week of year	Week of month x Month of year	End of month x Month of year
First stage F stat	45	14.4	18.2	55.5

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Appendix H: Robustness check: sample restrictions

Table H1: Effect of search time on moving within 2 years, with additional sample restrictions

	Inclusive	Removing special programs	Removing homeless	Removing disabled	Removing all	Removing 2020
log(Search time), fitted	-0.086*** (0.025)	-0.073** (0.026)	-0.071** (0.025)	-0.101*** (0.028)	-0.091** (0.032)	-0.082*** (0.024)
Num.Obs.	431991	376894	386208	299143	253711	375834
Year x PHA fixed effects	X	X	X	X	X	X
Individual controls	X	X	X	X	X	X
First stage F stat	45	39.4	40.2	33.3	27.9	47.9

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table H2: Effect of search time on exiting the program within 2 years, with additional sample restrictions

	Inclusive	Removing special programs	Removing homeless	Removing disabled	Removing all	Removing 2020
log(Search time), fitted	-0.057+ (0.031)	-0.045 (0.031)	-0.045 (0.032)	-0.060+ (0.033)	-0.051 (0.036)	-0.091** (0.032)
Num.Obs.	431991	376894	386208	299143	253711	375834
Year x PHA fixed effects	X	X	X	X	X	X
Individual controls	X	X	X	X	X	X
First stage F stat	45	39.4	40.2	33.3	27.9	47.9

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table H3: Effect of search time on moving to a lower poverty ZIP Code, with additional sample restrictions

	Inclusive	Removing special programs	Removing homeless	Removing disabled	Removing all	Removing 2020
log(Search time), fitted	0.062+ (0.038)	0.073+ (0.039)	0.054 (0.041)	0.091* (0.045)	0.085+ (0.047)	0.055 (0.038)
Num.Obs.	423381	369965	379382	293740	249712	368381
Year x PHA fixed effects	X	X	X	X	X	X
Individual controls	X	X	X	X	X	X
First stage F stat	44.8	39.2	40.3	33	28	47.9

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001